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Benchmarking Agent Workflows for Software Development

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Background and problem statement

Coding agents now write features in minutes, so the interesting work has moved to the process around them: how you plan, spec, test, and review. The community has built competing workflow “stacks” (spec-driven, test-first, vendor defaults, and no process at all [“vibe coding”]) and each claims to be the better way to work. Nobody has run the same tasks under several stacks and measured what actually changes.

Aim

Run one set of coding tasks under several agent workflows and measure what each one produces, at what cost. The main research question is: “How does the choice of agent workflow affect the quality of the code an agent produces?”

Objectives

- Identify various agentic workflows, from no process to spec-driven.
- Identify various coding tasks for benchmark, from an existing repository and on a new small project
- Run experiments across every task under every workflow.
- Compare and analyse the results.

Expected Outcomes

- A working test harness with run logs for every workflow.
- Reproducible scripts and results
- [Stretch goal] A publication

How will this opportunity benefit you?

Skill development in the following areas:

- Running coding agents with different setups (skills, hooks, worktrees).
- Designing fair experiments with controls and statistics.
- Measuring code quality and test coverage
- Reading and applying recent research.
- Clear writing, code review, version control.

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Other details

- Located on-campus at Burwood
- 10 weeks in duration (Unpaid)
- Undertaken as 2 days per week for a total of ~160 hours
- To be completed within the 12-week trimester period